DSA Final Theory Exam [Summer 2022]

Marks = 1000 **Time = 150 minutes**

Question 1: [200 Marks]

Consider a singly singled list with the **head pointer only** (a pointer to the tail is not available). Write down the code to print (display) this linked list in **reverse order** (from tail to head).

Question 2: [200 Marks]

Write down recursive codes for the following:

i. To reverse an input number integer. The reversed number should be returned by the value returning function.

ii. To find the n^{th} number of the following series. The user will enter the value of n:

`1, 2, 3, 6, 12, 23, 44, 85, 164, 316, …`

Question 3: [200 Marks]

We have a sorted doubly linked list (in ascending order) of distinct nodes (no two nodes have the same data) and a value x. We have to count triplets in the list that sum up to a given value x. A triplet is defined as values of those three nodes in a doubly linked list whose sum equals x. A function to perform the task of this question (in the class Doubly Linked List). No other function is required. An example is given below. You have to make a generic code; this means that the values (of DLL and x) can be different from the ones shown below:

DLL: 1 ↔ 2 ↔ 4 ↔ 5 ↔ 6 ↔ 8 ↔ 9
x = 17

Output:

Number of triplets = 2
The triplets are:
(2, 6, 9)
(4, 5, 8)

Question 4: [200 Marks]

Write down a function to find the sum of all those nodes in a Binary Search Tree (BST) whose value is greater than a given number N.

Question 5: [200 Marks]

Consider the values given below:

70	55	60	42	83	90	93	101	65	56

Using these values, create (draw on paper) a self-balancing tree (Red-Black or AVL tree) that you have studied in your DSA course. All values must be inserted in the given order (add 70, then 55, 60, 42, 83, 90, 93, 101, 65, and 56). All intermediate steps of insertion should be shown.